Chapter 4 Assignment

Simple Models of Walking and Running

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Abstract

This report covers Chapter 4 of the course. Part I comments on the four textbook examples: the rimless wheel (Example 4.3), the compass gait (Example 4.4), the SLIP simulation (Example 4.5) and the SLIP apex-to-apex map (Example 4.6). The supplied notebooks were run, the results are interpreted with the theory of hybrid and passive-dynamic locomotion, and sensitivity studies on slope, initial conditions, energy and touchdown angle are reported. Part II solves the Section 4.6 exercise: the one-dimensional hopper. The energy analysis is carried out, the hopping controller is implemented, and the autograder gives 13/13. The complete Python code and notebooks accompany the submission; the most important code is reproduced here.

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1 Overview and Reproducibility

The assignment consists of:

- **Examples** (code supplied; run and comment): Examples 4.3, 4.4, 4.5 and 4.6.
- **Section 4.6** (all exercises): the one-dimensional hopper.

All simulations use Drake 1.58.0 and the `underactuated` package 2026.9.8 in `/Users/leo/venvs/drake/bin/python` (NumPy 2.5.4, SciPy 1.18.1, Matplotlib 3.11.2). The notebooks were executed headlessly with `COLAB_TESTING=1`, which prevents the MeshCat visualizer from blocking.

Contents	Location
Example notebooks	`Ch-4/Exp4'3-*.ipynb`, `Exp4'4-*.ipynb`, `Exp4'5-6-slip.ipynb`
Section 4.6 notebook	`Ch-4/Exp4'6-one_d_hopper.ipynb`
Analysis scripts and figures	`Ch-4/exp4_analysis/`
This report	`Ch-4/Ch4_Assignment_Report/`

The Section 4.6 autograder result is

```
Total score is 13/13.
```

2 Example 4.3: The Rimless Wheel

2.1 Model and hybrid dynamics

The rimless wheel is a rigid wheel with $n = 8$ spokes, no rim, and a point mass m at the hip. The half-angle between adjacent spokes is $\alpha = \pi/8$, the radius is $l = 1$ m, and the wheel rolls down a ramp of slope γ. The stance phase is a simple pendulum:

$$\ddot{\theta} = \frac{g}{l}\sin\theta, \tag{1}$$

where θ is measured from the vertical and $\theta = 0$ is the unstable upright position. Between impacts the mechanical energy

$$E = \frac{1}{2}ml^2\dot{\theta}^2 + mgl\cos\theta \tag{2}$$

is conserved.

At each foot strike the new spoke becomes the stance leg. Angular momentum about the new contact point is conserved, while the collision is inelastic:

$$\dot{\theta}^+ = \dot{\theta}^- \cos(2\alpha). \tag{3}$$

The kinetic energy is multiplied by $\cos^2(2\alpha)$; for $\alpha = \pi/8$ this is $1/2$, i.e. half of the kinetic energy is lost at every step.

2.2 Return map, fixed point and stability

Combining the stance-phase energy balance with the collision map gives the step-to-step return map

$$\omega_{n+1} = \cos(2\alpha)\sqrt{\omega_n^2 + 4\frac{g}{l}\sin\alpha\sin\gamma}, \qquad \omega_n := \dot{\theta}_n^+. \tag{4}$$

A step is possible only if the wheel can pass the upright position; the threshold is

$$\omega_1 = \sqrt{2\frac{g}{l}\left[1 - \cos(\gamma - \alpha)\right]}. \tag{5}$$

The rolling fixed point is

$$\omega^* = \cot(2\alpha)\sqrt{4\frac{g}{l}\sin\alpha\sin\gamma} = \cot(2\alpha)\sqrt{D}, \qquad D := 4\frac{g}{l}\sin\alpha\sin\gamma, \tag{6}$$

and it is a valid gait only if $\omega^* > \omega_1$. Linearising the return map at ω^* gives

$$\left.\frac{d\omega_{n+1}}{d\omega_n}\right|_{\omega^*} = \cos^2(2\alpha) = \frac{1}{2} < 1, \tag{7}$$

so the rolling fixed point is locally stable (for $\alpha < \pi/4$).

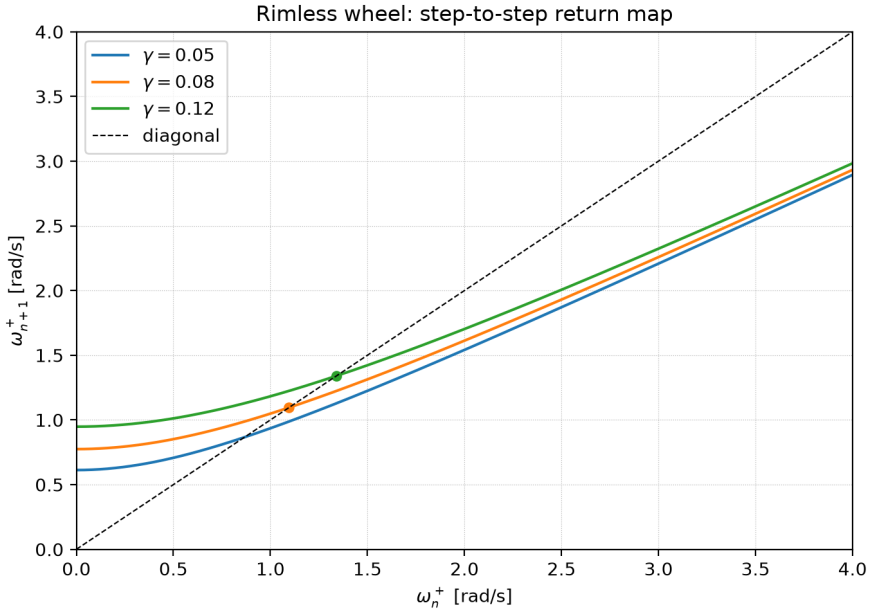
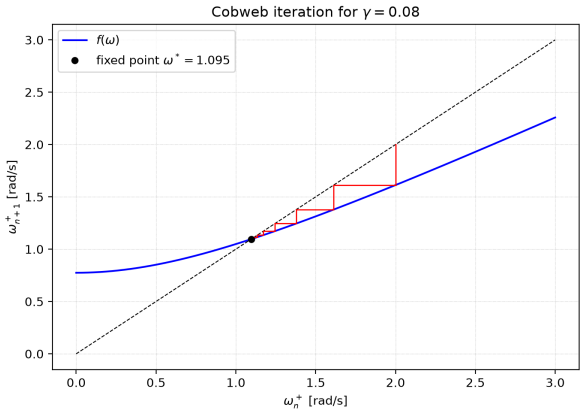


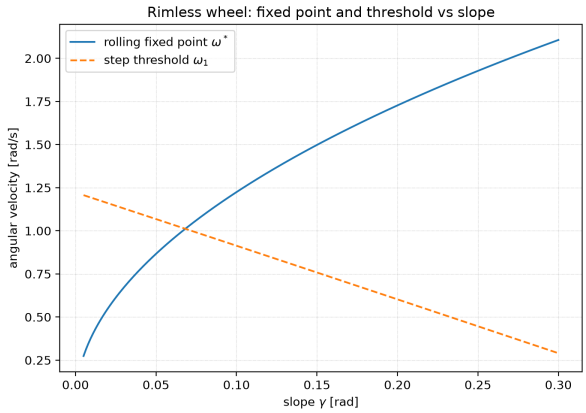
Figure 1: Rimless wheel return map for three slopes. Intersections with the diagonal are fixed points; only fixed points above the step threshold ω_1 correspond to a sustained rolling gait.

2.3 What was run and sensitivity study

The supplied notebook was run for initial angular velocities $5, 10, 0.95, -5, -4.8$ rad/s. Large positive velocities produce a rapid forward rolling gait; a small positive velocity stops before passing the upright position and swings back; negative velocities produce backward or oscillatory motion. This is exactly the three-branch structure of the return map in the lecture notes.



(a) Cobweb iteration at $\gamma = 0.08$.



(b) Fixed point and threshold vs. slope.

Figure 2: Rimless wheel: convergence to the stable fixed point and the existence condition $\omega^* > \omega_1$.

Table 1: Rimless wheel: threshold, fixed point and existence condition.

γ [rad]	ω_1 [rad/s]	ω^* [rad/s]	$\omega^* > \omega_1$?	$f'(\omega^*)$
0.02	1.1606	0.5480	No	0.5
0.05	1.0681	0.8663	No	0.5
0.08	0.9754	1.0955	Yes	0.5
0.12	0.8515	1.3408	Yes	0.5
0.20	0.6026	1.7272	Yes	0.5

Comments. The wheel is a passive-dynamic walker: there is no actuation. Gravity supplies energy during each stance phase, and the inelastic collision removes energy at each step. A stable periodic gait exists when these two effects balance, i.e. when the rolling fixed point lies above the step threshold. On very shallow slopes ($\gamma \lesssim 0.067$ rad in our parameters) the fixed point is below ω_1; the wheel cannot complete a step and eventually settles into the standing/oscillating branch. The stability multiplier is always $1/2$, independent of slope: the stronger dissipation at larger slopes is exactly compensated by the larger gravity input, so the linearised step-to-step dynamics remain equally stable. This is a clean example of a limit cycle generated by a hybrid system with impact dissipation.

3 Example 4.4: The Compass Gait

3.1 Model and hybrid dynamics

The compass gait consists of two point-mass legs of mass $m = 5$ kg and length $l = 1$ m ($a = b = 0.5$ m) connected at a hip of mass $m_h = 10$ kg. The minimal state is

$$\mathbf{x} = [\theta_{st}, \theta_{sw}, \dot{\theta}_{st}, \dot{\theta}_{sw}]^\top. \tag{8}$$

During stance the stance foot is pinned; during swing the swing leg moves freely; at heel strike an inelastic collision occurs and the roles of the two legs are swapped. The textbook model walks downhill on a ramp of slope $\gamma = 0.0525$ rad.

3.2 Results of the supplied notebook

The notebook starts from $[0, 0, 0.4, -2.0]^\top$ with zero hip torque and runs for 8 s. The phase portrait (left leg angle vs. angular velocity) approaches a closed orbit: the passive compass gait reaches a stable limit cycle. The impact at heel strike produces a visible jump in the velocity, while the downhill slope replenishes the lost energy during the stance phase.

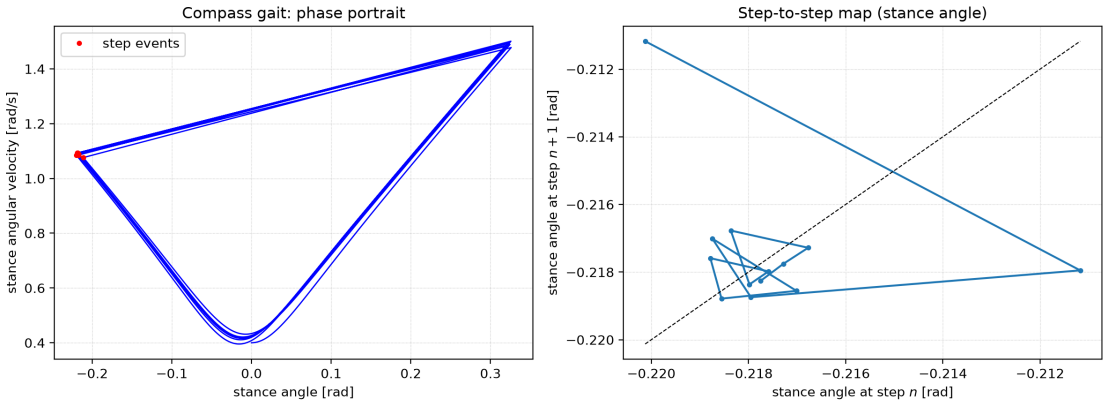


Figure 3: Compass gait: phase portrait (left) and step-to-step map of the stance angle (right). Red dots mark the detected heel strikes; the points converge to a fixed point of the step map.

3.3 Step detection and sensitivity

We detect a step by a change of the plant's abstract "left support" state. For the nominal initial condition the gait takes 14 steps in 10 s, with a mean step period of 0.735 s. The post-impact states converge rapidly to a neighbourhood of the limit cycle.

Table 2: Compass gait: initial-condition sensitivity (10 s horizon).

Initial state	Steps	Bounded/sustained?
$[0, 0, 0.4, -2.0]$	14	Yes
$[0, 0, 0.2, -1.5]$	7	No
$[0, 0, 0.6, -2.5]$	1	No
$[0, 0, 0.4, -1.0]$	10	No

For the slope sensitivity we used a 3×3 grid of initial states $\dot{\theta}_{st}(0) \in \{0.2, 0.4, 0.6\}$, $\dot{\theta}_{sw}(0) \in \{-2.5, -2.0, -1.5\}$ and ran 12 s for each slope. A run is classified as sustained when it has at least 10 steps, the last five step periods are consistent, and the state remains bounded.

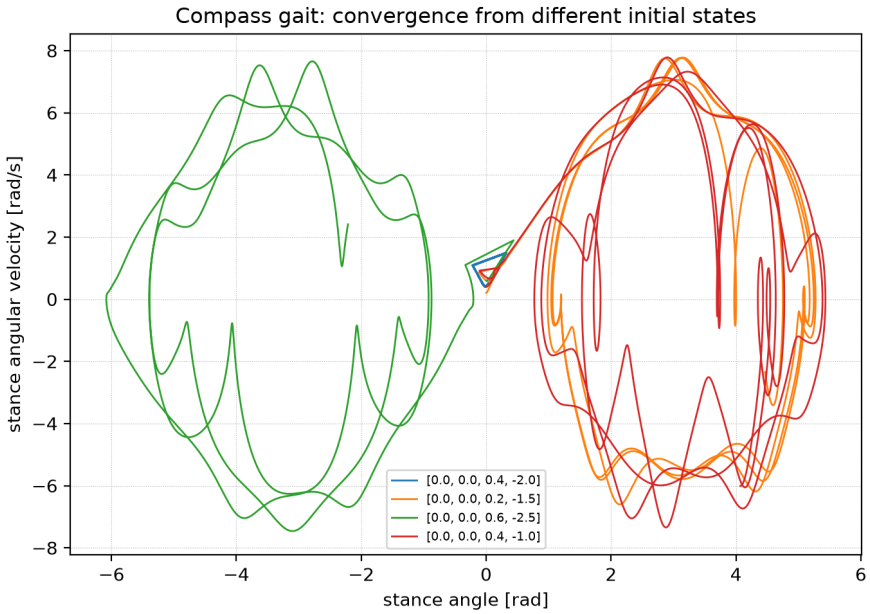


Figure 4: Compass gait from different initial states. Only a limited set of initial conditions reaches the limit cycle; others leave the narrow basin and fall.

Table 3: Compass gait: fraction of initial states leading to a sustained gait.

Slope γ [rad]	Successes / trials	Fraction	Mean period [s]
0.0200	3/9	0.33	0.705
0.0350	3/9	0.33	0.715
0.0525	3/9	0.33	0.730
0.0800	0/9	0.00	–
0.1200	0/9	0.00	–

Comments. The compass gait is also passive: no hip torque is used. Its limit cycle is an example of a stable periodic orbit of a hybrid system; the step-to-step map plays the role of the Poincaré map and its linearised eigenvalues (Floquet multipliers) determine orbital stability. In contrast to the rimless wheel, the basin of attraction is much narrower: only a small set of initial conditions, and a limited range of slopes, leads to sustained walking. This is consistent with the lecture notes: with only two legs there is no "extra spoke" to catch the body, so the swing-leg landing must be timed precisely. Real robots therefore need feedback (energy injection, foot placement, posture control) to enlarge the basin.

4 Example 4.5: The SLIP Model

4.1 Model

The spring-loaded inverted pendulum (SLIP) models running with a point mass $m = 80$ kg, a massless spring leg of rest length $r_0 = 1$ m and stiffness $k = 10.7\,mg/r_0$, and a touchdown angle

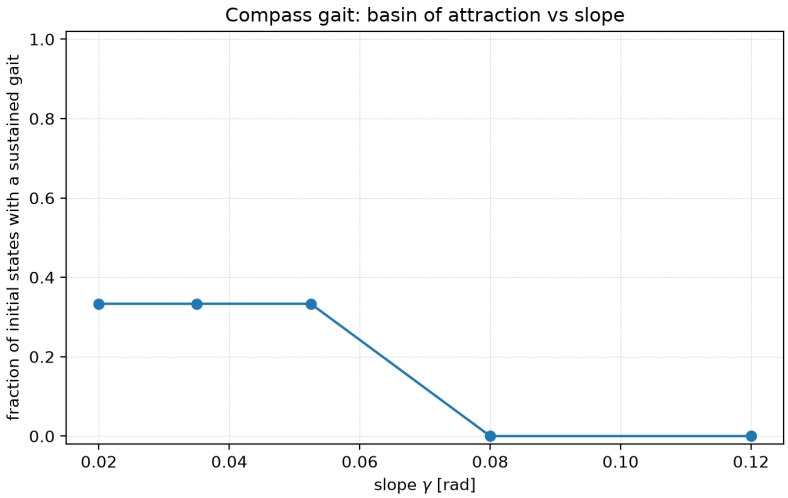


Figure 5: Compass gait: basin of attraction for the coarse initial-state grid as a function of slope.

that can be reset during flight. During flight the dynamics are ballistic:

$$\ddot{x} = 0, \qquad \ddot{z} = -g. \tag{9}$$

During stance, with r the leg length and θ the leg angle,

$$m\ddot{r} - mr\dot{\theta}^2 + mg\cos\theta - k(r_0 - r) = 0, \tag{10}$$

$$mr^2\ddot{\theta} + 2mr\dot{r}\dot{\theta} - mgr\sin\theta = 0. \tag{11}$$

The total energy

$$E = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) + mgr\cos\theta + \frac{1}{2}k(r_0 - r)^2 \tag{12}$$

is conserved: there is no impact loss because the leg and foot are massless. This is the key difference from the rimless wheel and compass gait.

4.2 Results of the supplied notebook

The notebook starts from a flight state with touchdown angle $\theta_{td} = 30^\circ$, dimensionless energy $\tilde{E} = 1.61$ and apex height $z = 0.9$ m, and simulates 2 s. The leg compresses during stance, stores elastic energy, and extends again; the body follows a running-like trajectory with a flight phase between stance phases. The notebook also builds the apex-to-apex map, which is analysed in the next section.

Comments. SLIP is a segmented Hamiltonian system. Unlike the walking models, it has no dissipative impact, so the energy is constant to integration accuracy. The stance phase is a spring-mass oscillator with centrifugal and gravity terms; the touchdown angle selects how much of the horizontal velocity is converted into compression. The model captures the essential running mechanism (compression and restitution) while remaining low dimensional.

5 Example 4.6: The SLIP Apex-to-Apex Map

5.1 Poincaré section and one-dimensional map

At the apex the vertical velocity is zero and the horizontal position is irrelevant because of translation symmetry. Energy conservation makes the horizontal velocity a function of the apex height,

$$\dot{x} = \sqrt{\frac{2}{m}\,(E - mgz)}, \tag{13}$$

so the return map on the constant-energy slice is one-dimensional:

$$z[n+1] = P(z[n]). \tag{14}$$

A fixed point $z^* = P(z^*)$ corresponds to a periodic running gait; its stability is determined by $P'(z^*)$ (stable if $|P'| < 1$).

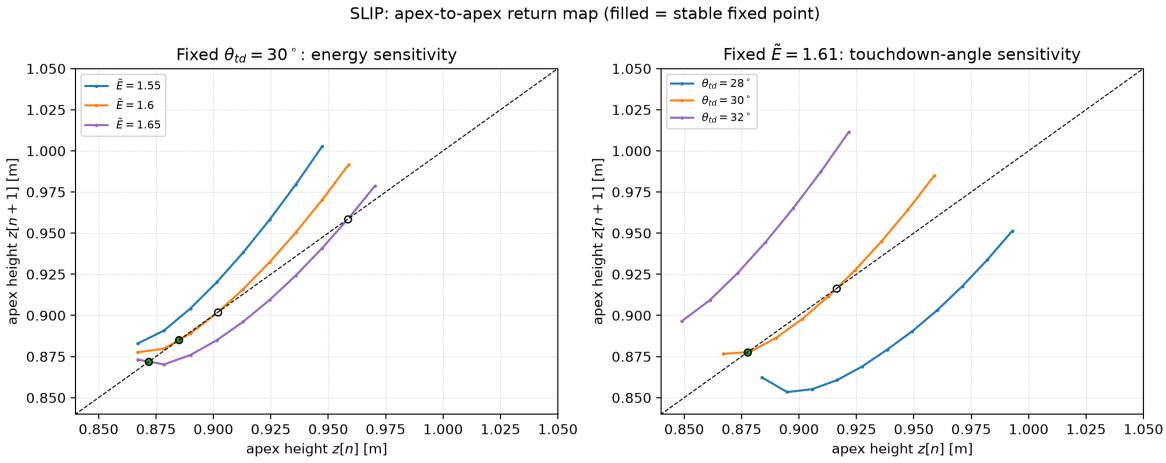


Figure 6: SLIP apex-to-apex map. Left: fixed touchdown angle $\theta_{td} = 30°$, varying energy. Right: fixed energy $\tilde{E} = 1.61$, varying touchdown angle. Filled circles are stable fixed points, open circles are unstable ones.

5.2 Fixed points and sensitivity

For the nominal condition $\theta_{td} = 30°$, $\tilde{E} = 1.61$ we find

$$z^*_{\text{stable}} = 0.8776, \quad P'(z^*) = 0.420, \qquad z^*_{\text{unstable}} = 0.9163, \quad P'(z^*) = 1.288. \tag{15}$$

These agree with the textbook values (0.8768 and 0.9168, with derivatives 0.488 and 1.346); the small differences come from the finite grid and linear interpolation.

Table 4: SLIP: fixed points for the tested parameter neighbourhood.

θ_{td} [deg]	$\tilde{E}$	z^* [m]	$P'(z^*)$
30	1.60	0.8850	0.500 (stable)
30	1.60	0.9019	1.167 (unstable)
30	1.61	0.8776	0.420 (stable)
30	1.61	0.9163	1.288 (unstable)
30	1.65	0.8719	0.122 (stable)
30	1.65	0.9586	1.513 (unstable)

Comments. The apex map has a stable and an unstable fixed point for the nominal parameters; initial heights between them converge to the lower (stable) gait, while heights above the unstable point leave the map's domain or fail. Changing the energy by ± 0.05 moves the fixed points but leaves a stable gait; changing the touchdown angle to 28° or 32° removes the fixed point in our scan. The stable fixed point at the nominal condition is therefore a narrow operating point, consistent with the map's limited domain.

Partial stability. The one-dimensional map is obtained by projecting the two-dimensional apex state $(z, \dot{x})$ onto z using energy conservation. Consequently, $|P'| < 1$ only proves stability with respect to perturbations that preserve the total energy. It does not prove stability to terrain changes, pushes or other disturbances that change the energy. Since SLIP is conservative, this is a *partial* stability result, not asymptotic stability of the full running motion. This distinction is important for control design: the apex map is useful for one-step-ahead controllers, but a real robot needs additional energy regulation.

6 Section 4.6: The One-Dimensional Hopper

6.1 Model and hopping cycle

The hopper has a body of mass m_b and a leg of mass m_l connected by an elastic prismatic joint of stiffness k. With z_b the body height and $l = z_l - z_b$ the leg compression, the state is $[z_b, l, \dot{z}_b, \dot{l}]^\top$. The spring force is $-k(l + \delta)$, where $\delta \geq 0$ is the preload control.

The cycle has four phases: flight, touchdown (inelastic impact of the leg), the stance phase (leg on the ground, body moving down and up), and lift-off (inelastic impact between body and leg). The controller changes δ only twice per cycle:

- at the bottom t_{bt}: set $\delta(t_{bt}^+) = \bar{\delta}$;
- at the apex t_{ap}: reset $\delta(t_{ap}^+) = 0$.

6.2 Energy derivation of the preload

The mechanical energy is

$$E = \frac{1}{2} m_b \dot{z}_b^2 + \frac{1}{2} m_l (\dot{z}_b + \dot{l})^2 + m_b g z_b + m_l g (z_b + l) + \frac{1}{2} k (l + \delta)^2. \tag{16}$$

Before the preload is applied at the bottom, $l = -z_b$ and $\dot{z}_b = 0$:

$$E(t_{bt}^-) = m_b g z_b + \frac{1}{2} k z_b^2. \tag{17}$$

After applying $\bar{\delta}$,

$$E(t_{bt}^{+}) = m_b g z_b + \frac{1}{2} k(\bar{\delta} - z_b)^2. \tag{18}$$

At lift-off, the body is at the hard stop and the leg is on the ground. Water energy conservation from bottom to lift-off gives the body kinetic energy before impact:

$$\frac{1}{2} m_b \dot{z}_b^2(t_{lo}^{-}) = m_b g z_b + \frac{1}{2} k z_b^2 - k\bar{\delta} z_b. \tag{19}$$

Momentum conservation for the inelastic lift-off impact gives

$$E(t_{lo}^{+}) = \frac{m_b}{m_b + m_l} \left[m_b g z_b + \frac{1}{2} k z_b^2 - k\bar{\delta} z_b \right] + \frac{1}{2} k \bar{\delta}^2. \tag{20}$$

At the apex both body and leg are at rest at height h:

$$E(t_{ap}^{-}) = (m_b + m_l) g h + \frac{1}{2} k \bar{\delta}^2. \tag{21}$$

Since energy is conserved during flight, $E(t_{lo}^{+}) = E(t_{ap}^{-})$, which gives the linear equation for $\bar{\delta}$:

$$\boxed{\bar{\delta} = \frac{1}{2} z_b + \frac{m_b g}{k} - \frac{(m_b + m_l)^2}{k m_b z_b} g h.} \tag{22}$$

If this value is negative, the system already has too much energy; the controller then uses $\bar{\delta} = 0$ and lets the impacts remove energy.

6.3 Implementation

Listing 1: Mechanical energy of the hopper.

```
def mechanical_energy(state, delta):
    k, mb, ml, g = get_hopper_parameters()
    zb, l, zb_dot, l_dot = state
    zl_dot = zb_dot + l_dot
    E = (0.5 * mb * zb_dot**2 + 0.5 * ml * zl_dot**2
         + mb * g * zb + ml * g * (zb + l)
         + 0.5 * k * (l + delta)**2)
    return np.asarray(E)
```

Listing 2: Preload controller.

```
class PreloadController(LeafSystem):
    def __init__(self, h):
        LeafSystem.__init__(self)
        self.DeclareVectorInputPort("state", 4)
        self.DeclareVectorOutputPort("delta", 1, self.DoCalcVectorOutput)
        self.h = h
        self.hopper_parameters = get_hopper_parameters()
        self.last_delta = 0.0
        self.last_state = np.full(4, np.nan)

    def is_bottom_or_apex(self, state):
        if np.any(np.isnan(self.last_state)):
```

```
            return None
        zb, l, zb_dot, l_dot = state
        last_zb_dot = self.last_state[2]
        if last_zb_dot <= 0.0 and zb_dot > 0.0:
            return "bottom" if zb < 0.0 else "apex"
        if last_zb_dot >= 0.0 and zb_dot < 0.0:
            return "apex" if zb > 0.0 else "bottom"
        return None

    def get_delta_bar(self, state):
        k, mb, ml, g = self.hopper_parameters
        zb = state[0]
        return (0.5 * zb + mb * g / k
                - ((mb + ml) ** 2) * g * self.h / (k * mb * zb))

    def DoCalcVectorOutput(self, context, delta):
        state = self.get_input_port(0).Eval(context)
        event = self.is_bottom_or_apex(state)
        if event == "bottom":
            delta.SetFromVector([max(self.get_delta_bar(state), 0.0)])
        elif event == "apex":
            delta.SetFromVector([0.0])
        else:
            delta.SetFromVector([self.last_delta])
        self.last_delta = delta[0]
        self.last_state = np.array(state)
```

6.4 Results and autograding

The notebook was executed end-to-end. The autograder reports

Total score is 13/13: mechanical energy 3/3, preload controller 10/10.

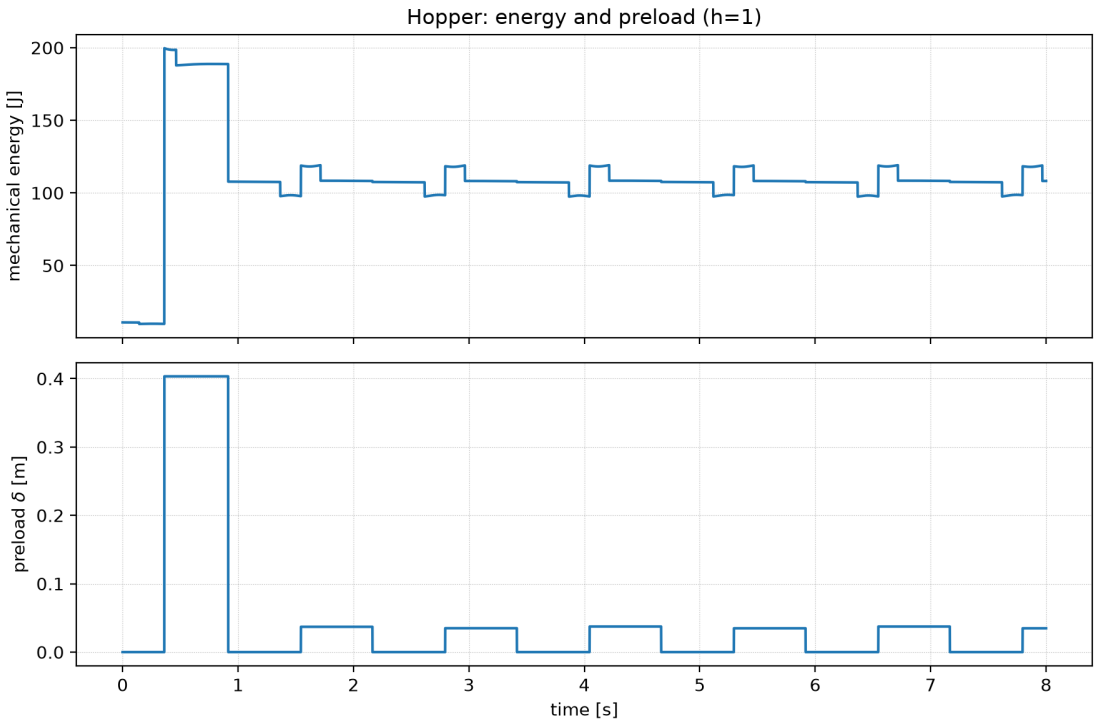


Figure 7: Hopper energy and preload for $h = 1$ m. The preload is applied at the bottom and reset at the apex; the energy drops at the two impacts.

Table 5: Hopper: target-height sensitivity (final 2 s of an 8 s run).

Target h [m]	Max body height [m]	Error [m]	Max preload δ [m]
0.50	0.4985	−0.0015	0.024
0.75	0.7477	−0.0023	0.093
1.00	0.9966	−0.0034	0.163
1.25	1.2444	−0.0056	0.232
1.50	1.4946	−0.0054	0.302

Table 6: Hopper: initial-height sensitivity for $h = 1$ m.

Initial $z_b(0)$ [m]	Max height (last 2 s) [m]	Error [m]	Max preload [m]
0.1	0.9966	−0.0034	0.403
0.3	0.9966	−0.0034	0.247
0.5	0.9966	−0.0034	0.163
0.8	0.9966	−0.0034	0.078
1.0	0.9966	−0.0034	0.037

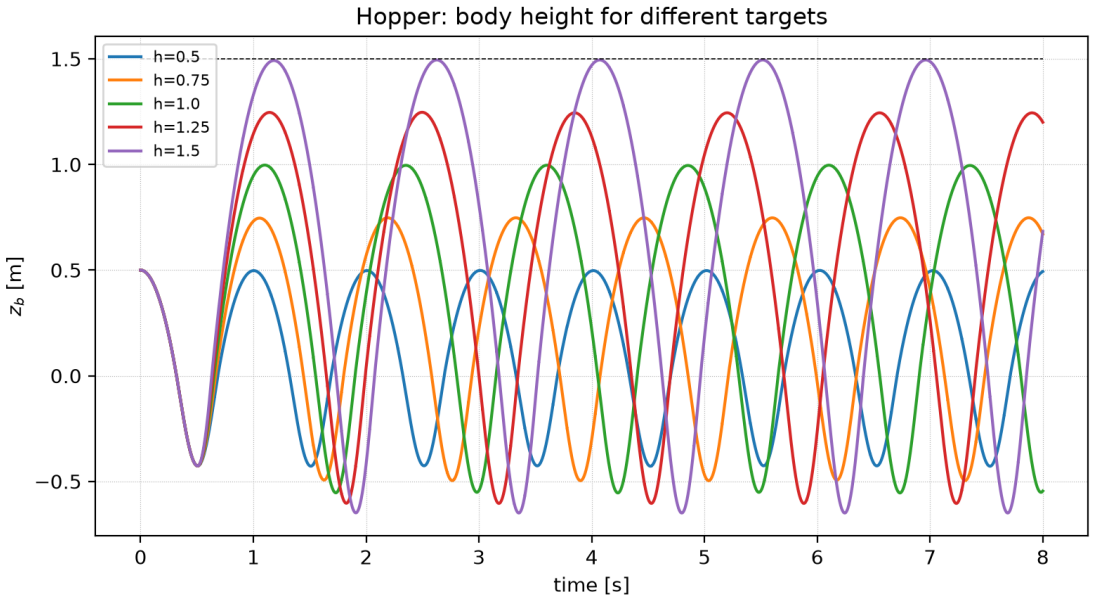


Figure 8: Hopper: body height for different target heights. After a short transient all runs settle into a periodic hop at the requested height.

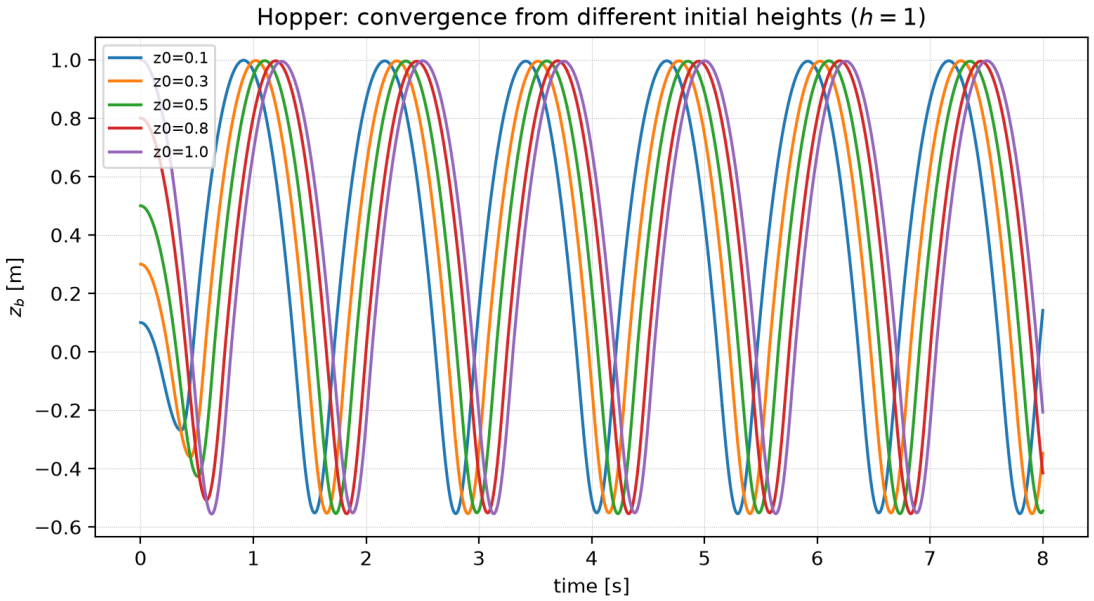


Figure 9: Hopper: convergence from different initial body heights at $h = 1$ m. The energy controller rejects the initial-height error.

Comments. The controller is an energy-based one-step controller: it does not track a reference trajectory, but chooses the preload exactly so that the next apex has the desired mechanical energy. The target-height error is below 6 mm for all tested heights, and the controller is insensitive to the initial body height. The maximum preload grows roughly linearly with the target height: a higher hop needs more energy, and the spring stores it as $\frac{1}{2}k\bar{\delta}^2$. The initial transient is larger

when the hopper starts low, because more energy must be injected at the first bottom; after that the cycle is periodic. This is a clean example of using a Poincaré/energy argument to reduce a trajectory-tracking problem to a scalar control decision once per hopping cycle.

7 Conclusions

1. **Passive dynamic walking.** The rimless wheel and the compass gait show that a stable periodic gait can arise without actuation: gravity supplies energy during stance and collisions remove it at foot strike. The step-to-step return map is the natural tool; its fixed point is the gait and its slope/Floquet multipliers determine orbital stability.

2. **Rimless wheel.** The return map is one-dimensional and the fixed point is stable with multiplier $\cos^2(2\alpha) = 1/2$. The main sensitivity is the slope: below a critical slope the fixed point lies below the step threshold and no rolling gait exists.

3. **Compass gait.** The limit cycle is much more delicate: only a narrow set of initial conditions and slopes leads to sustained walking. The coarse sensitivity grid confirms the small basin of attraction; feedback is needed for robustness.

4. **SLIP.** The model is conservative and captures running as a sequence of flight and spring stance phases. The apex map has a stable and an unstable fixed point for the nominal parameters, but the one-dimensional stability is only partial because it relies on energy conservation. The touchdown angle and energy strongly affect whether a fixed point exists.

5. **Hopper.** The energy analysis gives a closed-form preload $\bar{\delta}(z_b, h)$; the controller applies it at the bottom and resets at the apex. The autograder passes 13/13, the hopping height error is below 6 mm, and the controller is robust to the initial body height.

8 Code and Reproduction

The complete code is in the repository. Main entry points:

Topic	Script / notebook
Example 4.3	`Exp4'3-rimless-wheel.ipynb`, `rimless_wheel_analysis.py`
Example 4.4	`Exp4'4-compass-gait.ipynb`, `compass_gait_analysis.py`
Examples 4.5/4.6	`Exp4'5-6-slip.ipynb`, `slip_analysis.py`
Section 4.6	`Exp4'6-one_d_hopper.ipynb`, `one_d_hopper_analysis.py`

Commands (from `Ch-4/exp4_analysis`):

```
/Users/leo/venvs/drake/bin/python rimless_wheel_analysis.py
/Users/leo/venvs/drake/bin/python compass_gait_analysis.py
/Users/leo/venvs/drake/bin/python slip_analysis.py
/Users/leo/venvs/drake/bin/python one_d_hopper_analysis.py
```

Headless notebook execution:

```
COLAB_TESTING=1 /Users/leo/venvs/drake/bin/jupyter nbconvert \
  --to notebook --execute --inplace --ExecutePreprocessor.timeout=300 \
  "Exp4'6-one_d_hopper.ipynb"
```

The report is compiled with

```
pdflatex Ch4_Report.tex
pdflatex Ch4_Report.tex
```

A Key Code Snippets

A.1 Rimless wheel return map

```
def f_map(omega, gamma):
    return np.cos(2 * ALPHA) * np.sqrt(
        np.maximum(omega**2 + D(gamma), 0.0))

def fixed_point(gamma):
    return (1.0 / np.tan(2 * ALPHA)) * np.sqrt(D(gamma))
```

A.2 SLIP apex map

```
def apex_map(touchdown_angle, Etilde, n=30, zmax=1.2):
    plant = SpringLoadedInvertedPendulum()
    s = SLIPState(np.zeros(8))
    s.theta = touchdown_angle
    s.r = 1.0
    simulator = Simulator(plant)
    context = simulator.get_mutable_context()
    plant.get_input_port(0).FixValue(context, [touchdown_angle])
    context.SetAccuracy(1e-5)
    zs = np.linspace(np.cos(touchdown_angle) + 0.001, zmax, n)
    zns = np.full(n, np.nan)
    for i, z in enumerate(zs):
        s.z = z
        s.xdot = plant.apex_velocity_from_dimensionless_system_energy(Etilde, z)
        context.SetTime(0.0)
        context.get_mutable_continuous_state_vector().SetFromVector(s[:])
        simulator.Initialize()
        simulator.AdvanceTo(0.6)
        if plant.last_apex is not None:
            zns[i] = plant.last_apex
        plant.last_apex = None
    return zs, zns
```